Protected LiPo -> fused battery -> modem switch and 3.3V LDO -> ESP32 + switched ePaper

USB-C is data/programming only. Charge battery externally. Design remains a prototype.

Global net labels connect functional sheets. See DESIGN_REPORT.md for sequencing and validation gates.

Power

File: power.kicad_sch

Controller

File: controller.kicad_sch

Modem

File: modem.kicad_sch

Interfaces

File: interfaces.kicad_sch

Display

File: display.kicad_sch

Sensors

File: sensors.kicad_sch

Not released for fabrication; bench and RF validation required

Hector / xmyoot

Sheet: /

File: Low-Power_Cellular_ePaper_Device_ESP32-S3_SIM7600.kicad_sch

Title: Cellular ePaper - revision B architecture

Size: A3	Date: 2026-10-21	**Rev: B / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 1/7

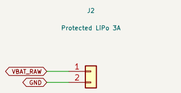
Protected 1S LiPo, external charging only. USB does not power or charge this board.

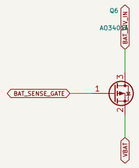
Battery connector: pin 1 positive, pin 2 GND. Verify pack polarity before connection.

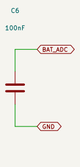
AP2112K: 600mA maximum, 55uA typical quiescent current. No 10uA system claim.

Disable modem before loaded battery falls below 3.6V; verify rail stays above 3.4V.

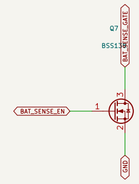
Battery ADC: enable divider, wait >=250ms before sampling (RC = 50ms).

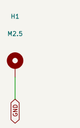


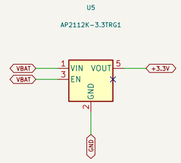


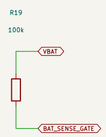








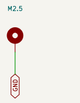


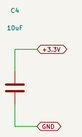












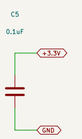














Not released for fabrication; bench and RF validation required

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Sheet: /Power/

File: power.kicad_sch

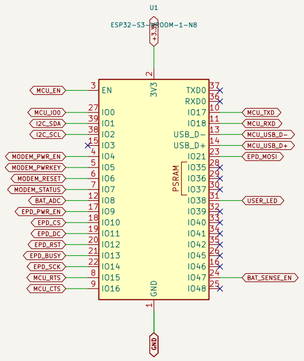
Title: Power

Size: A2	Date: 2026-10-21	**Rev: B / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 2/7

ESP32-S3-WROOM-1-N8: 8MB flash, no PSRAM. GPIO47 enables battery divider.

USB-C data/programming only. Protected battery must be connected.

Do not drive display SPI/reset high while VCC_EPD is off; disable internal pull-ups.

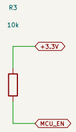








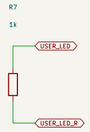


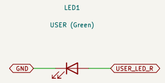


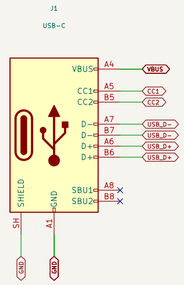








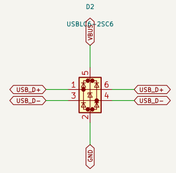


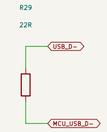


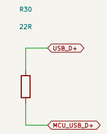












Not released for fabrication; bench and RF validation required

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Sheet: /Controller/

File: controller.kicad_sch

Title: Controller

Size: A2	Date: 2026-10-21	**Rev: B / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 3/7

SIM7600G-H 135-pad package, hardware design V1.07 table 5 and figure 4.

VBAT 3.4-4.2V, design for 2A transmit pulses. Validate battery and transient response.

Assert PWRKEY with N-FET for >=500ms; reset pulse 100-500ms.

Use AT+CPOF and wait for STATUS low before switching modem power off.

Unused and reserved module terminals explicitly NC; DTR held low for UART awake.

U2

SIM7600G-H

GND	1	GND	RI	69	
GND	2	GND	DCD	70	
PWRKEY_1V8	3	PWRKEY	TXD	71	MODEM_TXD_1V8
RESET_1V8	4	RESET	DTR	72	GND
GND	5	GND	PCM_OUT	73	
	6	SPI_CLK	PCM_IN	74	
	7	SPI_MISO	PCM_SYNC	75	
	8	SPI_MOSI	PCM_CLK	76	
	9	SPI_CS	GND	77	GND
GND	10	GND	GND	78	GND
MODEM_USB_VBUS	11	USB_VBUS	GNSS_ANT	79	RF_GNSS
MODEM_USB_D-	12	USB_DN	GND	80	GND
MODEM_USB_D+	13	USB_DP	GND	81	GND
GND	14	GND	MAIN_ANT	82	RF_MAIN
VDD_EXT	15	VDD_1V8	COEX1	83	
	16	USB_ID	COEX2	84	
USIM_DATA	17	USIM_DATA	USB_BOOT	85	
USIM_RST	18	USIM_RST	COEX3	86	
USIM_CLK	19	USIM_CLK	GPIO77	87	
USIM_VDD	20	USIM_VDD	RESERVED	88	
	21	SD_CMD	RESERVED	89	
	22	SD_DATA0	RESERVED	90	
	23	SD_DATA1	RESERVED	91	
	24	SD_DATA2	RESERVED	92	
	25	SD_DATA3	GND	93	GND
	26	SD_CLK	GPIO4	94	
	27	SDIO_DATA1	GPIO2	95	
	28	SDIO_DATA2	GPIO1	96	
	29	SDIO_CMD	GPIO54	97	
	30	SDIO_DATA0	GND	98	GND
	31	SDIO_DATA3	RESERVED	99	
	32	SDIO_CLK	RESERVED	100	
	33	GPIO3	RESERVED	101	
	34	GPIO6	RESERVED	102	
	35	HSIC_STROBE	RESERVED	103	
	36	HSIC_DATA	RESERVED	104	
GND	37	GND	RESERVED	105	
VBAT_MODEM	38	VBAT	DBG_RXD	106	
VBAT_MODEM	39	VBAT	RESERVED	107	
GND	40	GND	SGMII_RST_N	108	
GND	41	GND	SGMII_INT_N	109	
	42	DBG_TXD	GND	110	GND
GND	43	GND	RESERVED	111	
	44	VDD_AUX	SGMII_RX_P	112	
	45	ISINK	SGMII_RX_M	113	
	46	ADC2	SGMII_TX_M	114	
	47	ADC1	SGMII_TX_P	115	
	48	SD_DET	SGMII_MDIO	116	
MODEM_STATUS_1V8	49	STATUS	SGMII_MDC	117	
	50	GPIO43	GND	118	GND
	51	NETLIGHT	GPIO19	119	
	52	GPIO41	GND	120	GND
	53	USIM_DET	GND	121	GND
	54	FLIGHTMODE	GND	122	GND
	55	SCL	GND	123	GND
	56	SDA	GND	124	GND
GND	57	GND	GND	125	GND
GND	58	GND	GND	126	GND
RF_AUX	59	AUX_ANT	GND	127	GND
GND	60	GND	GND	128	GND
GND	61	GND	GND	129	GND
VBAT_MODEM	62	VBAT	GND	130	GND
VBAT_MODEM	63	VBAT	GND	131	GND
GND	64	GND	GND	132	GND
GND	65	GND	GND	133	GND
MODEM_RTS_1V8	66	RTS	GND	134	GND
MODEM_CTS_1V8	67	CTS	GND	135	GND
MODEM_RXD_1V8	68	RXD			

Q2

BSS138

MODEM_PWR_EN → 1 (G) · 3 (D) → MODEM_PWR_GATE · 2 (S) → GND

Q3

BSS138

MODEM_PWRKEY → 1 (G) · 3 (D) → PWRKEY_1V8 · 2 (S) → GND

Q4

BSS138

MODEM_RESET → 1 (G) · 3 (D) → RESET_1V8 · 2 (S) → GND

Q1

AO3400

MODEM_PWR_GATE → 1 (G) · 3 (D) → MODEM_PWR · 2 (S) → VBAT

R8

100k

VBAT — MODEM_PWR_GATE

R9

10k

MODEM_PWR_EN — GND

R27

100k

MODEM_PWRKEY — GND

R28

100k

MODEM_RESET — GND

C11

220uF 6.3V T520D227M006ATE015

VBAT_MODEM — GND

C12

220uF 6.3V T520D227M006ATE015

VBAT_MODEM — GND

C13

47uF 10V

VBAT_MODEM — GND

C14

47uF 10V

VBAT_MODEM — GND

C15

10uF

VBAT_MODEM — GND

C16

10uF

VBAT_MODEM — GND

C17

0.1uF

VBAT_MODEM — GND

C18

0.1uF

VBAT_MODEM — GND

C19

33pF

VBAT_MODEM — GND

C23

100nF

PWRKEY_1V8 — GND

C24

100nF

RESET_1V8 — GND

PWR_FLAG

VBAT_MODEM

Not released for fabrication; bench and RF validation required

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Sheet: /Modem/

File: modem.kicad_sch

Title: Modem

Size: A2	Date: 2026-10-21	**Rev: B / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 4/7

TXS0104E OE is referenced to the 1.8V VCCA supply.

U7 translates STATUS to 3.3V; DIR tied to VCCA selects A→B.

Main and diversity LTE antennas required. GNSS connector is passive (no DC bias).

RF matching is provisional; use fabricator stackup and verify 50-ohm routes.

J9: external modem USB service, pin1 VBUS, pin2 D−, pin3 D+, pin4 GND.

U3
TXS0104EQ
+1.8V
+3.3V
1
14
VCCA
VCCB
OE
A1
A2
A3
A4
B1
B2
B3
B4
GND
8
2
3
4
5
13
12
11
10
7
UART_OE
MODEM_TXD_1V8
MODEM_RXD_1V8
MODEM_RTS_1V8
MODEM_CTS_1V8
MCU_RXD
MCU_TXD
MCU_RTS
MCU_CTS
GND
U7
SN74LVC1T45DBVR
+1.8V
+3.3V
1
6
VCCA
VCCB
A
B
DIR
GND
3
4
5
2
MODEM_STATUS_1V8
MODEM_STATUS
VDD_EXT
GND
C21
0.1uF
VDD_EXT
GND
C22
0.1uF
+3.3V
GND
C38
100nF
VDD_EXT
GND
C39
100nF
+3.3V
GND
C30
100nF
USIM_VDD
GND
R13
22R
USIM_CLK
USIM_CLK_SIM
R14
22R
USIM_DATA
USIM_DAT_SIM
R15
22R
USIM_RST
USIM_RST_SIM
D9
PESD3V3U1UL
USIM_RST_SIM
GND
J4
MAIN_ANT_UFL
RF_MAIN_CONN
1
2
GND
R16
0R
RF_MAIN
RF_MAIN_CONN
C31
DNP
RF_MAIN
GND
C32
DNP
RF_MAIN_CONN
GND
C33
47pF
RF_GNSS
RF_GNSS_CONN
J8
LTE diversity
RF_AUX
1
2
GND
J9
Modem USB service
MODEM_USB_VBUS
MODEM_USB_D−
MODEM_USB_D+
GND
1
2
3
4
J5
GNSS_ANT_UFL
RF_GNSS_CONN
1
2
GND
J3
nanoSIM_GCT_SIM8060
USIM_VDD
USIM_RST_SIM
USIM_CLK_SIM
GND
GND
USIM_DAT_SIM
GND
1
2
3
4
5
6
7
8
VCC
RST
CLK
GND
VPP
I/O
SH
PWR_FLAG
MODEM_USB_VBUS
D6
PESD3V3U1UL
USIM_CLK_SIM
GND
D7
PESD3V3U1UL
USIM_DAT_SIM
GND
R11
10k
VDD_EXT
UART_OE
R24
100k
UART_OE
GND
C29
1uF
USIM_VDD
GND

Not released for fabrication; bench and RF validation required

Hector / xmyoot

Sheet: /Interfaces/

File: interfaces.kicad_sch

Title: Interfaces

Size: A2	Date: 2026-10-21	**Rev: B / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 5/7

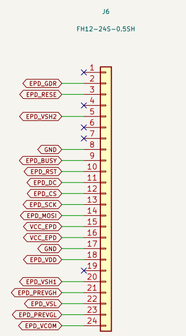
GDEY029T94 2.9 inch 296x128 monochrome, four-wire SPI (BS1 grounded).

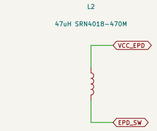
Panel pin18 VDD is an internal rail: capacitor only, never connect to 3.3V.

Reference: GDEY029T94 specification, page30 (PDF page29), typical application.

Q9 SI1308EDL: L2 47uH >=500mA; flying capacitor 4.7uF/25V; sense 2.2 ohm.

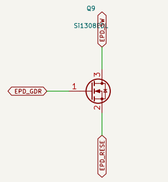
Verify FPC contact side and pin1 with panel mechanical drawing before ordering.





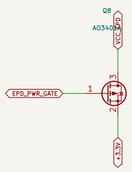










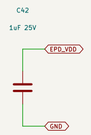


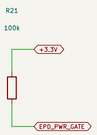






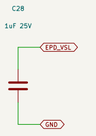




















Not released for fabrication; bench and RF validation required

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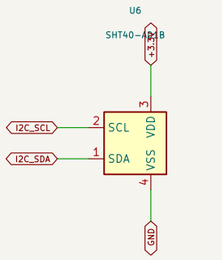
Sheet: /Display/
File: display.kicad_sch

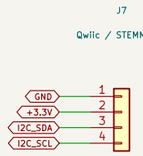
Title: Display

Size: A2	Date: 2026-10-21
KiCad E.D.A. 10.0.7	**Rev: B / PROTOTYPE**
	Id: 6/7

SHT40 and external Qwiic share 3.3V I2C. Pull-ups 4.7k.

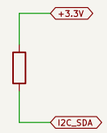
Place sensor at board edge, away from modem, LDO and display boost heat.

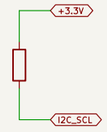












Not released for fabrication; bench and RF validation required

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Sheet: /Sensors/
File: sensors.kicad_sch

Title: Sensors

Size: A3	Date: 2026-10-21	**Rev: B / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 7/7